ADITYA COLLEGE OF ENGINEERING AND TECHNOLOGY
No 2/9, Kamakshipura, Hesaraghatta Hobli, Yelahanka, Bengaluru- 560089
Affiliated to VTU, Belagavi | Approved by UGC/AICTE/Govt. of Karnataka, Bengaluru

DEPARTMENT OF PHYSICS

ASSIGNMENT – III

Academic Year	2024-25	**Dept.**	Physics
Program	BE	**Date of issue**	23-06-2025
Year / Semester /Sec	2025/II/A/B/C	**Date of Submission**	28-06-2025
Course Code	**BPHYS202**		
Course Name	Applied Physics for CSE Stream		

Note	***1. Answer all the questions***

Q. No.	Questions	Marks	COs	RBT level
1	Enumerate the failures of classical free electron theory and discuss the assumptions of quantum free electron theory of metals.	10	CO4	L3
2	Define Fermi Energy level. Discuss the various energy states by the electrons at T = 0 K and T > 0 K on the basis of fermi factor.	10	CO4	L3
3	Calculate the probability of occupation of an energy level 0.2 eV above fermi level at temperature 27 ^{0}C.	05	CO4	L3
4	Describe Meissner effect.	07	CO4	L3
5	Elucidate SQUIDS and Josephson's Effect.	08	CO4	L3
6	Describe in detail about Type- I and Type- II Superconductors.	08	CO4	L3
7	Define critical temperature and critical magnetic field. Explain briefly BCS theory of superconductivity.	07	CO3	L3
8	Elucidate the importance of Size & Scale, Weight and strength in animations	05	CO5	L2
9	Discuss the timing in Linear motion, Uniform Motion, ease in (Slow in) and ease out (slow out) in animation.	10	CO5	L2
10	Illustrate the odd rule and odd rule multipliers with a suitable example.	08	CO5	L3
11	Describe Jumping, parts of Jump and Jump Magnification.	08	CO5	L2
12	Distinguish between descriptive and inferential statistics.	05	CO5	L2
13	Elucidate Poisson distribution with an example.	08	CO5	L3
14	Slowing-in object in an animation has a first frame distance 0.5m and the first slow in frame 0.35m. Calculate the base distance and the number of frames in sequence.	05	CO5	L3

Prepared by:

Prof. Nayan Kumar H N
Assistant Professor
Dept. of Physics

Prof. Pavendhan M
Associate Professor
HOD